Lab 3 — Component Modelling & Architectural Pattern Selection

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Municipal Infrastructure Damage Reporting App

Domain: Smart Cities, Transport & Logistics

Actors: Citizen Reporter, Municipal Engineer

Chosen Architecture: Microservices

Architecture Selection

We chose the Microservices Architecture for the Municipal Infrastructure Damage Reporting App.

Justification

Reason 1 — Independent Scaling for Variable, Unpredictable Load:

Photo-heavy GPS/EXIF extraction and report submission traffic spike sharply after storms, civic drives, or during peak commute hours, while report-routing and engineer-portal usage stay comparatively steady. A microservices style lets the GPS Location Extraction Component and Report Submission Component be scaled independently of the Report Management and Routing services, so infrastructure is added only where the load actually occurs (FR-001, FR-002).

Reason 2 — Fault Isolation for a High-Priority Offline Requirement:

NFR-001 requires offline drafting with auto-sync within 30 seconds of reconnection — a feature that must never block the core reporting path. Deploying Offline Sync as its own independent service means a failure or slow response there cannot take down report submission, GPS extraction, or routing; each service can fail and recover on its own without a single point of failure for the whole system.

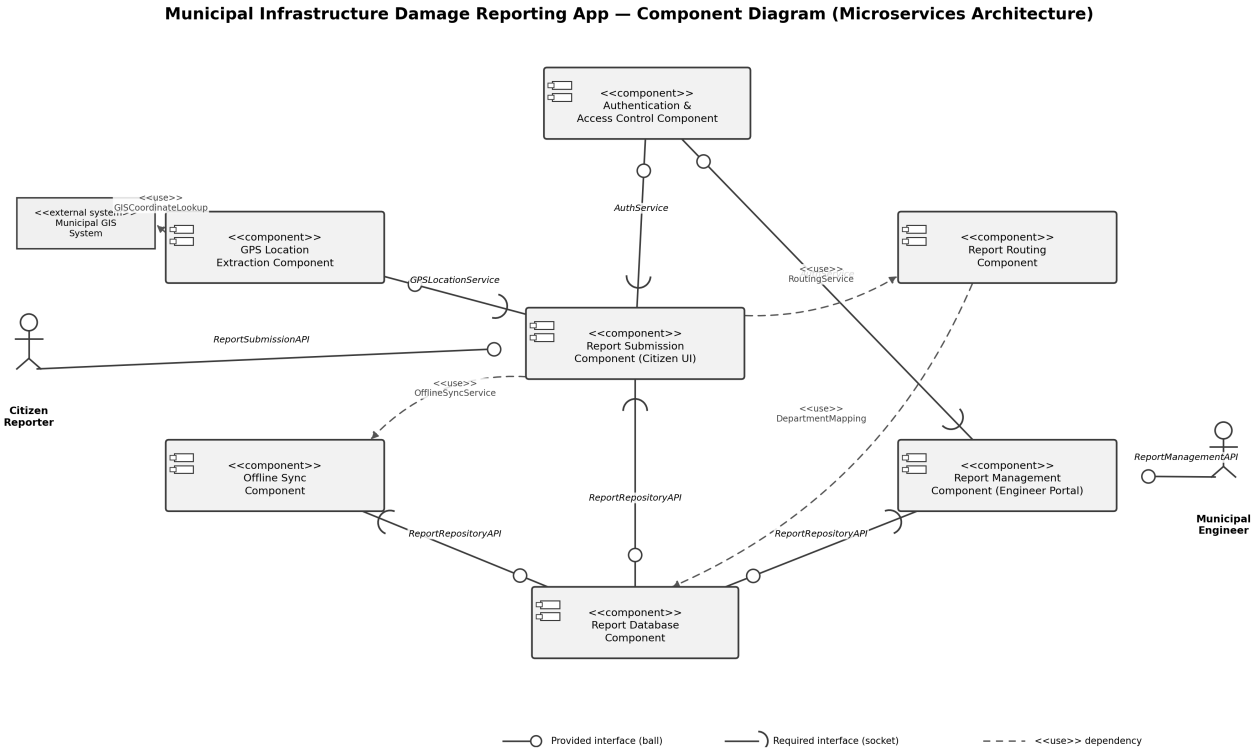
Security Advantage

A dedicated Authentication & Access Control Component centralises identity and authorization and is called independently by both the Report Submission Component and the Report Management Component before any read or write reaches the Report Database Component. This directly satisfies NFR-002 (restricting report modification to the original Citizen Reporter or an authorized Municipal Engineer): because every service must pass through this component rather than embedding its own ad-hoc checks, access rules are enforced consistently and an unauthorized modification attempt is rejected at the interface level, independent of which client or service initiated the request.

Performance Benefit

Because each microservice is independently deployable, the GPS Location Extraction Component can run on image-processing-optimised infrastructure and be scaled up specifically for EXIF metadata parsing without over-provisioning the rest of the system. This keeps photo-upload latency low for citizens (supporting FR-001) while the Report Routing and Report Management services continue to run at normal capacity, giving better overall throughput than a single monolithic layer would under the same load.

UML Component Diagram



Components identified:

- Report Submission Component (Citizen UI) — orchestrates report creation, validation, and submission (given equivalent role)
- GPS Location Extraction Component — extracts EXIF GPS metadata from uploaded photos (FR-001)
- Report Routing Component — routes each report to the correct municipal engineering department (FR-003)
- Authentication & Access Control Component — enforces who may view/modify a report (NFR-002)
- Report Management Component (Engineer Portal) — lets a Municipal Engineer view and update report status (FR-004)
- Offline Sync Component — supports offline drafting and auto-sync on reconnect (NFR-001)
- Report Database Component — stores reports, statuses, and location data